LexGraph — Technical Report

Leakage, Discretisation and Authority Grounding in Indian Supreme Court Judgment Prediction

CSE 573: Semantic Web Mining — Arizona State University

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Abstract

This is the single authoritative document for the project. It supersedes and replaces the specification (`PROJECT.md`), the chronological development log (`WORKLOG.md`) and the nine stage-acceptance reports (`reports/*.md`), all of which were removed; their content is folded in here and their history remains in git. A plain-language companion for readers without a background in law or machine learning, `LexGraph_Explained.tex`, sits in this same folder.

Two results are established. **First**, the input construction used by the published state of the art on this task beats a facts-only input by $+0.111$ to $+0.181$ AUROC, and essentially all of that gap is disposition language surviving the deletion of the operative order. The residue extends roughly **128 words** past the boundary — a figure found independently on four splits — and a word-count-matched window set back past it scores no better than the facts on any of them. Across 26,688 judgments the effect is significant in all eight decades from the 1950s to the 2020s. The court's reasoning does carry real signal, but only about $+0.025$ AUROC: the apparent effect is four to seven times its true size.

Second, projecting extracted facts into a symbolic legal vocabulary costs 0.097 AUROC, which is most of what survives extraction. This is a property of discretisation rather than of any particular ontology: a corpus-induced vocabulary of equal granularity is statistically indistinguishable from the hand-authored one ($P = 0.846$), more labels do not help, and the cost is concentrated precisely where the vocabulary *succeeds* in naming a fact.

Third, ğ17 rebuilds the *intended* architecture — the one this project set out to construct — correctly, and localises where it loses information. Done properly, fact families lose nothing (0.633 from 345 dimensions against 0.629 for the 47,354-feature text they derive from). The element layer, by contrast, reaches 0.521 and fails its acceptance test, and three explanations for that were tested before the right one survived: the cause is the *basis*, not the encoding. At matched granularity a data-driven basis scores 0.632 where the 66-element doctrinal basis scores 0.524. Naming all 66 induced clusters shows why: legal-substance facts alone, procedural facts alone and dates alone all score ≈ 0.605, so **the outcome signal is diffuse rather than concentrated**, a spanning basis captures it and a selective doctrinal probe cannot, and no ontology would unlock it. ğ17.7 addresses whether the published 78% can be beaten: trivially yes on the leaky task, almost certainly not on the honest one — a fine-tuned InLegalBERT reaches 0.695 AUROC on facts (three seeds) against 0.907 on the contaminated window, the same model and training. ğ17.5 checks the element layer's verdicts against a blind AI reviewer: positive verdicts agree

80% of the time, defeat verdicts under half, and part of that weakness traces to court conclusions that extraction recorded as bare facts.

Neither critique is new to the literature, and ğ18 positions both against verified prior work. What is defensibly ours is methodological, and ğ18 says which parts. Three constructive results stand alongside: statute prediction at 1.9× the strongest baseline, precedent retrieval at 7.4× a popularity control, and a faithfulness-tested trace mechanism. Every number here is read from a file under `experiments/`.

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1 How this document is organised, and the specification section map

The project was built against a written specification whose section numbers (ğ5.2, ğ6.2, ğ10.1, ...) are cited in code comments across **50 source files**. That specification file has been removed in favour of this report, so those references are preserved here: each stage below carries its original spec number in the heading, and the table maps them.

Spec ğ	Stage	Here
3.1	Voyager/LLM environment	ğ2
4	Corpus reconnaissance	ğ3
5.1	Outcome labels	ğ4
5.2	Masking — the only text a predictor may see	ğ5
5.3	Splits	ğ6
6.1	Fact extraction	ğ7
6.2	Canonical vocabulary	ğ8
6.4	Gold annotation set	ğ21
7	Claim families	ğ9
8	Fact patterns	ğ10
8.3	Element burden metadata	ğ21
9.1	Statute normalisation	ğ11
9.2	Statute prediction	ğ11
9.3	Precedent retrieval	ğ12
10.1	Outcome models and feature groups	ğ13
10.2	Ablations	ğ13
11	Traces and faithfulness	ğ14

Results that are not in the specification at all are in ğ15 (the leakage measurement that became the project's main finding) and ğ17 (the intended architecture, rebuilt correctly and measured, whose code lives in `v2/`). ğ19 is the development record: the defects found and what each would have caused. ğ20 lists claims held and withdrawn. ğ21 is what is *not* established.

2 Environment spec ğ3.1

All model calls go to ASU's hosted endpoint (Voyager, `https://openai.rc.asu.edu/v1`), through the single door in `src/llm/client.py`. No paid API is used anywhere.

- **No VPN required.** The endpoint resolved to private `10.139.64.x` historically; as of 2026-09-18 it resolves to public Cloudflare addresses and answers from a residential connection.
- **Generation**: `llama4-scout-17b` by default. **Embeddings**: `qwen3-embedding-8b` (4096-dim). Only three of the roughly forty-nine advertised models serve `/v1/embeddings`; the rest 404 there, and the model list carries no capability metadata, so probing is the only way to tell an embedder from a generator.
- **Content-addressed cache**, keyed `sha256(model, messages, params)` in SQLite. Re-running an experiment is free — the LLM baseline ran at a 100% hit rate over 600 calls. The cache holds 162,230 entries.

- `ok=False` **means "never got an answer"**, which is deliberately distinct from a model that replied with nothing. Conflating the two turned a network outage into a finding twice before this was enforced (ğ19).
- **90-second timeout.** The client default of 600 s × 3 retries froze an early batch for over twenty minutes with no output.
- **Parallelism**: 96 workers for extraction, 24 for embeddings; throughput went from 10.3 to 38.6 documents per second.

Setup: `.venv/bin/pip install -r requirements.txt`, then `.env` at the project root with `VOYAGER_API_KEY=<key>` from `voyager.rc.asu.edu` → LLM Access → Create Key.

3 Corpus spec ğ4

26,688 Indian Supreme Court judgment PDFs, 1950–2025, 6.8 GB, held locally in `Data/raw_pdfs/<year>/`. A provided spreadsheet identifies **6,954** of them as land or property disputes; that subset is the working corpus for ğ4–ğ14, and ğ15.3 drops the restriction and uses all 26,688.

Extracting the archives. Two of the four source zips contain filenames with invalid UTF-8 bytes. APFS rejects those names, `unzip` misreports it as `write error (disk full?)`, then blocks on a `Continue? (y/n)` prompt — and on non-interactive stdin takes EOF and **silently aborts the whole archive**, exiting 0. That is how a first attempt produced 12,422 of 26,688 files and reported success. Extraction must go through Python's `zipfile` with each name sanitised (`encode("utf-8","replace")`, NFC-normalise, strip control characters), and the count verified afterwards.

Joining spreadsheet rows to PDFs. `src/data/adapter.py` joins on the exact filename the spreadsheet supplies: **6,954 of 6,954 (100%)**, asserted at import time against `MIN_PDF_JOIN_RATE = 0.99` so a regression halts the pipeline instead of silently shrinking the dataset. An earlier normalised-title join achieved 0.6%, because judgment filenames carry a copy marker after the year (`..._on_23_May_1957_1`) and titles do not — so stripping "trailing digits" from both removes the *year* from every title and every pair mismatches by exactly that year.

Text. 6,954 of 6,954 extracted, zero failures, zero near-empty. Median 5,302 words.

What the dataset readme claims and the PDFs do not have. The readme describes structured sections. Sampled across six decades, `FACTS` appears as a header in **0** of 12 judgments, `ISSUE` in 2, `ARGUMENT` in 2, `REASONING` in 0. There is nothing to segment on, which is why ğ5 classifies units by cue rather than by heading.

Citation metadata. The provided "precedents" lists mix genuine case-to-case citations with statute and article references; only about **32.5%** are real case citations. The classifier output used downstream is `Data/processed/citations_classified.jsonl`. The metadata is Indian Kanoon's link graph and is independently scraped, not ground truth — English and Privy Council authorities are systematically absent. Separately, only **11,991 of 46,904** case-citation edges (25.6%) point at a case that is in this corpus at all, because the corpus is Supreme Court only while judgments cite High Courts, the Privy Council and English decisions. Recall must therefore be reported against *reachable* authorities with the raw figure beside it.

4 Labels spec ğ5.1

Two independent labellers, made to agree.

Rules (`src/data/labels.py`) locate the operative order by regular expression and classify it. The hard part is that Indian orders dispose of several proceedings in one breath — "the appeal is allowed … and the writ petition stands dismissed" — where the second clause is a *consequence* of the first and a naive matcher reads it as the outcome. Proceedings are therefore ranked into tiers: an appeal or special leave petition (`_PROC_T1`) governs, and a writ petition, suit, revision or application (`_PROC_T2`) is consulted only when tier 1 is silent. That change alone cut rules-versus-LLM conflicts from 196 to 107. A bounded gap (`_GAP`) forbids a proceeding noun from binding across another disposition verb, and a negator check rejects "we do not think it proper to again remand the matter".

LLM sees the same order window and must quote the span it relied on.

Agreement: 87.0% on the 4,317 cases where both decided and the quote verified. Conflicts are held as `UNRESOLVED` and queued in `Data/interim/label_conflicts.jsonl` rather than resolved by trusting one labeller.

Label	Cases	Share
LOSE	2,964	42.6%
WIN	2,365	34.0%
PARTIAL	624	9.0%
UNRESOLVED	562	8.1%
UNKNOWN	289	4.2%
REMAND	116	1.7%
OTHER	34	0.5%
Total	**6,954**	

Table 1: The binary task uses WIN and LOSE only: 5,329 cases.

5 Masking spec ğ5.2

`src/data/mask.py` keeps only material that existed before the court decided. Units are classified by cue, not by heading, because there are no headings (ğ3).

Section	Kept	Why
Caption, parties, judges	no	Judge identity predicts but is not a legal signal; party names let a model memorise repeat litigants.
Facts	**yes**	The genuine input.
Procedural history	*isolated*	Legitimately known in advance but a shortcut, so it goes to its own channel `prior_court_text` and feature group `Q`, never mixed in silently.
Arguments of counsel	**yes**	Written by the parties, before the decision.
Court's reasoning	no	Written after, knowing the answer.
Operative order	no	This is the label.
HEADNOTE (pre-2000)	no	An editorial summary that states outcome and authorities outright.
Page furniture	no	Indian Kanoon URLs, page numbers, running headers.

Within kept units there is a second, sentence-level scrub: sentences matching `OUTCOME_CUE` are dropped, and sentences matching `PRIOR_COURT_DISPOSITION` are diverted to the shortcut channel. Average retained length is about 2,500 words of the original 5,300.

Three variants are produced per case, which is what makes ğ15 possible: `masked_text` (facts and pleadings only), `ildc_style_text` (everything but the order) and `ildc_style_no_headnote`.

The sensitivity control, and why everything else depends on it. A claim to have removed leakage is untestable on its own. So a deliberately leaky arm is built — a model trained on *only* the deleted text — and must score near-perfectly. If it does not, the order detector is broken and the masked arm cannot be trusted either.

Input	Acc.	Macro-F1	AUROC
Majority class	0.528	0.345	0.500
Prior court's disposition only	0.562	0.562	0.585
Masked text	0.629	0.626	0.674
Whole unmasked judgment	0.746	0.745	0.827
The removed order only	**0.931**	**0.931**	**0.981**

Table 2: Leakage probe, `forum_heldout`, $n_{\text{test}} = 902$. `experiments/paper3/leakage_probe_forum_heldout.jsc`
An LLM version agrees: 95.7% accuracy on the order alone, 62.0% on masked text ($n = 350$).

6 Splits spec ğ5.3

A random split fails here twice over. The same dispute generates several judgments (appeal, review, connected matter), so groups must not be split; and training on later cases to predict earlier ones is time travel.

- `forum_heldout` — three forums of origin held out entirely. Train 3,350 / dev 455 / test 902. Hash `6e346dd75b7e7d84`.

- `temporal_2004_2013` — train $\leq$2004, dev 2004–2013, test >2013. Train 3,446 / dev 480 / test 768. Hash `43d9d645667ad3a5`.

Every results file records the split hash, so a number traces to the exact split that produced it.

A conflict a unit test caught. Grouping by a case group's *maximum* year put 1970 cases in "test >2013"; by *minimum* year it put 2025 cases in "train $\leq$2004". Both are time travel, and no assignment satisfies both constraints for a group straddling the boundary. The 13 straddling groups are dropped and the count recorded. `tests/test_stage1.py` enforces it.

7 Fact extraction spec ğ6.1

A model reads each segmented unit and proposes *facts*: atomic assertions, each carrying a verbatim quote, who asserted it, and whether it was disputed. `src/grounding.py` locates the quote in the source; a fact whose quote cannot be found is **discarded, not corrected**.

Grounding rate 0.92. Current scale: **96,623 facts across 4,702 cases**, median 17 per case, 15,377 claim records.

Segmentation. The first implementation split on blank lines and returned *one unit* for 57.7% of cases, because this PDF text frequently contains no blank lines. A sentence-window tier was added: median units per case 1 $\rightarrow$ 47, unusable cases 57.7% $\rightarrow$ 2.6%.

The verifier was the bug, not the extractor. An early version rejected 42% of quotes that were in fact correct: PDF extraction inserts hyphenation artifacts (`plain- tiff's`), and a 20-character floor rejected true short quotes such as "Appeal dismissed." (15 characters). After repairing hyphenation and lowering the floor, the verified rate on a hand-checked set went from 58.3% to 91.7%, with 15 of 16 confirmed present by eye.

Model selection. Six models on an identical 20-case set:

Model	Ground.	Leak	Attrib.	Facts/case	s/case
`llama4-scout-17b`	0.92	0.03	0.27	18.4	**4.2**
`llama4-maverick-17b`	**0.93**	**0.01**	**0.41**	**23.5**	28.7
`qwen3-30b-a3b-instruct-2507`	0.92	0.03	0.40	10.5	15.7
`olmo3-32b-instruct`	excluded — about 50 s per call				
`glm-5-3-flash`	excluded — hangs on long prompts				
`glm-5-3`	excluded — too slow at corpus scale				

Table 3: `experiments/paper1/extractor_benchmark.json`. `maverick` is better on every quality axis and 6.8$\times$ slower; `scout` was chosen for corpus-scale runs and the trade-off is kept on the record.

Attribution, and an unexplained ceiling. Party attribution sits at **0.19–0.20**. The model records admittedness in `disputed_status` rather than `asserted_by`, and reading both raised coverage from 17% to 96% — but the *rate* at which a fact is attributed to a party has not moved. A prediction that recovering wrongly-deleted argument text would fix it was tested and refuted (ğ20). The cause is unknown.

8 Canonical vocabulary spec ğ6.2

Each fact is mapped from free text to a controlled label drawn from a 345-entry vocabulary generated from the ontology's named fields. `ontology/ontology_v1.yaml` defines 15 claims, 66 elements, 6 defence families and 4 remedy tiers.

The appeal of a controlled vocabulary is comparability across cases: you can count element satisfaction, mine patterns, and reason symbolically. The cost is that two distinct facts mapping to one label become indistinguishable — and ğ16 shows that cost is severe.

Three failures worth not repeating. (i) The canonicaliser's self-reported "NEW" rate of 0.7% was untrustworthy: 16.7% of facts were out of vocabulary and unflagged. Out-of-vocabulary is now decided by checking the answer, not the flag; at scale the model flags only 75 of 3,155 (2.4%). (ii) Batching with a shared union menu wrecked label fidelity, and out-of-vocabulary rose with menu size (51%/75%/76% at batch 4/8/16); per-fact menus restored parity (32.3% vs 34.4% unbatched). (iii) Repointing `PROMPT_ID` left `canonicalize_one` validating a batched reply against a single-object schema, so it returned `None` for everything — and it is the *fallback* path. Split into `PROMPT_ID_ONE` and `PROMPT_ID`.

9 Claim families spec ğ7

Clustering 7,441 claims (HDBSCAN and agglomerative, `src/cluster/`) produced six clusters with a **silhouette of 0.0245** and 71.5% of points classed as outliers; NMI 0.237, ARI 0.083. A silhouette near zero means the clusters are not separated. Merging with a minimum family size let one family (`PrivateTitlePossession`) absorb 2,365 cases and nearly every other label.

The taxonomy is not supported by the data, and is reported as such rather than presented as a finding.

10 Fact patterns spec ğ8

FP-Growth over closed itemsets within each family, chi-square tested, Benjamini–Hochberg corrected, with bootstrap stability over 20 resamples.

Closed itemsets mined	498
Stable across $\geq$80% of 20 bootstraps	311
Patterns statistically tested	351
Pass uncorrected $p < 0.05$	10
Expected by chance alone	*17.6*
Survive Benjamini–Hochberg	**0**

Ten patterns passed an uncorrected test — *fewer* than the 17.6 chance predicts. Without the correction those ten would have shipped as a pattern library. This is the clearest case in the project of a control changing the conclusion, and it follows directly from ğ16: pattern mining over symbols that carry no signal cannot find anything. The specification's ğ8 deliverable is therefore **vacuous rather than met**.

11 Statutes spec ğ9.1, 9.2

`src/authorities/normalize.py` reduces citations to canonical `CODE:provision` form over a registry of 42 Acts. Three properties of real Indian citations forced the design:

- **Act names arrive truncated.** The most frequent "Act" in 1,200 sampled judgments is *Property Act* (644 mentions) — the *Transfer of Property Act* with its head cut off by a line break. *India Act* (516) is the *Government of India Act*. Truncations resolve through a path tagged `resolved_by="truncated"`, so a guess is never reported as a full-name match.
- **Generic back-references dominate**: *the Act*, *the said Act* (311+), which resolve only by looking back to the last Act actually named, within a bounded window.
- **Implicit Acts**: a bare *Article* means the Constitution; an *Order* or *Rule* means the Civil Procedure Code.

Coverage over 1,500 judgments: 58,295 citations, **61.4% resolved**, median 6 distinct provisions per case, 149 cases with none.

A copy control killed the first result. Predicting the provisions that apply reached micro-F1 0.447 and looked strong. A system that does nothing but copy provisions already named verbatim in the input scores **0.713** — far higher. The "prediction" was largely string matching. The honest version removes every provision the input mentions, leaving only those that must be inferred:

System	Micro-F1	P@1	R@10
Global top-5 prior	0.069	0.134	0.418
Claim-family prior *(strongest baseline)*	0.093	0.173	0.503
Copy from text *(control, now at floor)*	0.000	0.081	0.385
Masked text	**0.173**	**0.236**	**0.660**
Extracted fact text	0.159	0.206	0.581
Canonical atoms	0.109	0.161	0.464

Table 4: Novel provisions only. 46 labels, $n_{\text{test}} = 702$. `experiments/paper2/statute_prediction_forum_heldout_labels_novel.json`.

Micro-F1 0.173 against 0.093 — $1.9\times$ **the strongest baseline**, with the copy control at zero by construction. A real positive result, and note the ordering recurs: raw text $>$ extracted facts $>$ canonical atoms.

12 Precedents spec ğ9.3

150 queries. Recall is against the **1,035 of 2,113** cited authorities (49.0%) that are themselves in this corpus (ğ3); the raw figure is stated beside it. Every retrieved authority is asserted to pre-date the query judgment — checked per result, not trusted to a filter.

System	R@10	R@50	MRR	nDCG@10
Mention baseline	0.022	0.112	0.037	0.014
Popularity *(control)*	0.032	0.103	0.072	0.032
Canonical atom overlap	0.021	0.058	0.046	0.021
Dense, mean-pooled	0.053	0.130	0.114	0.054
Dense, max-pooled	0.039	0.104	0.097	0.044
BM25	0.207	0.350	**0.431**	0.225
Dense over masked text	0.229	0.390	0.380	0.221
RRF(BM25 + dense masked)	**0.238**	**0.453**	0.428	**0.243**

Table 5: `experiments/paper2/precedent_retrieval_forum_heldout.json`.

R@10 0.238 and MRR 0.431 against a popularity control of 0.032 and 0.072 — about 7.4× **on R@10.** Because the control is popularity, beating it by that margin establishes the system responds to the facts rather than exploiting citation skew.

Two sub-results: canonical atom overlap (0.021) falls *below* the popularity control, so the symbolic representation fails here too; and max-pooling made the dense arm *worse* (0.039 vs 0.053), a prediction that had to be withdrawn.

13 Outcome models spec ğ10.1, 10.2

Six explicitly separated feature groups, so any claim about what drives a prediction is testable by ablation. 1,415 features: `F` canonical fact atoms 1,313; `P` mined patterns 3; `E` element satisfaction **0 (empty, see ğ21)**; `S` predicted statutes 49; `R` retrieved precedents 6; `C` case metadata 38; `Q` prior court's disposition 6.

Model	Acc.	Macro-F1	95% CI	ECE
Majority (global)	0.530	0.346	[0.332, 0.360]	—
Majority (per decade)	0.571	**0.557**	[0.527, 0.589]	—
`lr F` (facts only)	0.501	0.499	[0.464, 0.530]	0.270
`lr F+P+E+S+R+C`	0.513	0.513	[0.479, 0.546]	0.285
`lr S only`	0.599	0.599	[0.567, 0.629]	**0.041**
`lr Q only`	0.525	0.509	[0.474, 0.540]	0.034
`gbm C only` (metadata)	0.583	0.582	[0.550, 0.613]	0.091
`gbm F+P+E+S+R+C`	**0.598**	**0.597**	[0.566, 0.630]	0.107

Table 6: $n_{\text{test}} = 881$, 24 models run. `experiments/paper3/outcome_forum_heldout.json`.

Three uncomfortable observations. **Every confidence interval spans about 0.11 and they all overlap** — the best structured model (0.597) and a per-decade majority baseline (0.557) are not distinguishable at this sample size. **Adding facts to metadata does not help**: `gbm C only` reaches 0.582 on year and forum alone, while facts alone reach 0.517, and McNemar confirms the metadata-and-authority groups genuinely beat the fact group ($P = 0.0008$ for LR, $P = 0.004$ for GBM) — a real difference in the wrong direction for the project's original thesis. And **calibration and accuracy come apart**: `lr S only` is the best-calibrated arm at competitive accuracy.

LLMs do no better. Five configurations on the same 300 cases: `LLM-0` 0.491 macro-F1, `LLM-FS` 0.522, `LLM-CoT` 0.582, `LLM-CF` 0.469, `LLM-CF+A` 0.503, against structured GBM 0.599 and TF-IDF on masked text 0.593. Chain-of-thought's edge over zero-shot is not significant (McNemar $P = 0.926$). Nothing clears a per-decade majority baseline by a resolvable margin, so the ceiling is not an artifact of a weak model family.

The ceiling looks intrinsic. 354 errors on 881 cases were categorised with a **base-rate control**, and it is decisive. Against an overall error rate of 0.402 the lifts are `family_unassigned` 1.12, `statute_mismatch` 0.99, `discretionary` 0.93, `extraction_thin` 0.88, `atoms_only_generic` 0.62. **Every lift is approximately 1.0** — not one of the pipeline's own quality signals predicts where it will fail. Reported as raw counts ("175 errors involve a statute mismatch") they would have read as a diagnosis and a work plan.

The stronger test: on 300 shared cases the structured model made 119 errors and `LLM-CoT` 125; under independence one expects $119 \times 125/300 = 49.6$ wrong for both. Observed: **46**. The two systems' errors are statistically independent, so they are not failing for a shared, fixable reason.

14 Traces spec ğ11

A deletion test. Remove the facts the trace cites, recompute features *from scratch*, and measure how far the predicted probability moves; compare against deleting the same number of random facts, and the facts the trace ranked least important. Deletion is applied to the **facts**, not the feature vector — deleting a feature the model weights heavily would be circular.

Facts deleted	Mean Δp	vs cited	95% CI
Cited by the trace	**0.0650**	—	—
Random	0.0209	+0.0441	[+0.0245, +0.0659]
Inverse-ranked	0.0180	+0.0470	[+0.0291, +0.0666]

Table 7: 45 cases. Cited facts move the prediction 3.1× as much as random, winning on 77.8% of cases. `experiments/paper3/trace_faithfulness_forum_heldout_gbm.json`.

Verdict: faithful. Faithful is not correct — the test shows the explanation describes the model's actual computation, not that the computation is legally sound. Given that the same model predicts near chance, a faithful explanation of a near-chance prediction is an honest account of a weak inference.

15 The leakage result

This is not in the specification. It became the project's main finding.

15.1 The construction

Six input constructions on **identical cases, split, labels and classifier** (TF-IDF into logistic regression). Only the text the model may read varies.

Input	Words	Acc.	Macro-F1	AUROC
`order_only` *(control)*	427	0.921	0.920	0.976
`full` *(ceiling)*	5,302	0.720	0.718	0.788
`ildc_style` *(all but the order)*	5,051	0.680	0.678	0.734
`ildc_style_no_headnote`	4,481	0.678	0.676	0.735
`last_512_tokens`	**511**	**0.776**	**0.775**	**0.845**
`masked` *(facts + arguments)*	2,501	0.621	0.619	0.668

Table 8: 895 test cases, 3,324 training. `experiments/paper1/reasoning_ablation_forum_heldout.json`. Paired bootstrap: `last_512−masked` $= +0.1764$ $[+0.1480, +0.2064]$; `ildc_style−masked` $= +0.0660$ $[+0.0539, +0.0789]$.

`last_512_tokens` reaches **accuracy 0.776**; the published best on this task is **78%**. A plain TF-IDF model lands within half a point of a transformer result once it reads the same window. Note also that the winning arm reads **511** words against masked's **2,501** — so "less text" cannot explain the gap; it runs hard the other way.

15.2 Where the advantage actually comes from

The final 512 tokens sit exactly where the operative order used to be. If the boundary is even slightly imperfect, that arm — and no other — is contaminated. Three measurements, every arm exactly 512 words so volume is held constant.

1. The setback curve. Four non-overlapping windows stepping back from the end. Both predicted shapes were written into the module before it ran: *residual leakage* predicts a cliff after the first window, because contamination cannot reach back 512 words; *reasoning* predicts gradual decay with every window above `masked`.

It is unambiguously the cliff (table in ğ15.3, all four splits). **512 words of the court's reasoning taken from anywhere but the very end carry no more outcome signal than the facts do**, and on the corpus-wide runs slightly less.

2. The mechanism, visible in the text. Indian judgments announce the conclusion one sentence *before* the operative order:

> "For the foregoing reasons we find no merit in the appeal. *The appeal is dismissed with costs.*"

The detector marks the boundary at the second sentence, which is **correct** — that is the operative order. The first sentence stays, and states the outcome just as plainly.

3. Depth. Sweeping the window back in fine steps locates the boundary, and AUROC falls together with the disposition-cue rate — what a single shared cause looks like. On `forum_heldout`: offset 0 gives AUROC 0.838 at cue rate 0.419; 32 words 0.780/0.296; 64 words 0.719/0.208; **128 words 0.684/0.131, no longer significant**; 256 words 0.676/0.099; 512 words 0.669/0.089.

What cue-scrubbing cannot do. Deleting every cue-*bearing sentence* from the final window drops it 0.838 → 0.775; evaluating only on the 489 cases with no detected cue gives 0.742 against `masked`'s 0.646 on those same cases. Both remain above `masked`, so sentence-level scrubbing does **not** fully clean the window — there is telegraphing language the patterns miss ("we find no merit in the submission", "the conclusion is inescapable"). This is exactly why the setback curve is the primary evidence: it never relies on detecting a cue.

15.3 Generalisation: 26,688 judgments, every case type

Two objections survive the above — it might be a land-dispute artifact, and 842 test cases is not many. The intended answer was a replication on ILDC (34,816 cases), whose release is gated; it was eventually run (ğ15.4). The corpus-scale test below came first and stands on its own, because **the restriction to land disputes was ours, not the data's**. All 26,688 judgments were already on disk and ILDC is drawn from the same population (Indian Kanoon's Supreme Court judgments). `src/data/corpus_wide.py` extracts, rules-labels and masks every one in parallel, API-free: **2.8 minutes**, yielding **14,168 usable WIN/LOSE cases** (12,067 usable in every arm) across criminal, tax, constitutional, service, company and land matters.

This is a *generalisation test, not a replication*: it reproduces neither ILDC's labels nor ILDC's deletion boundary.

	land only		**all case types**	
	`forum` *(842)*	`temporal` *(722)*	`temporal` *(2,288)*	`random` *(2,414)*
`masked`	0.656	0.577	0.628	0.723
`w0` (last 512)	0.838	0.716	0.739	0.862
`w1`	0.669	0.589	0.606	0.717
`w2`	0.647	0.577	0.616	0.715
`w3`	0.638	0.578	0.603	0.701
`w0−masked`	+0.181	+0.139	+0.111	+0.139
`w1−masked`	+0.013	+0.012	−0.022	−0.006
significant?	*no*	*no*	*no*	*no*
Advantage gone at	**128 w**	**128 w**	**128 w**	**128 w**

Table 9: Four independent splits. `leakage_setback_*.json`, `corpus_scale_setback_*.json`.

Not a pre-2000 reporting artifact. The pre-2000 HEADNOTE convention is a plausible alternative cause. It is not the cause — `w0−masked` is significant in **every decade**:

decade	1950s	1960s	1970s	1980s	1990s	2000s	2010s	2020s
n	112	438	377	292	278	338	349	230
Δ	+0.179	+0.325	+0.186	+0.180	+0.171	+0.163	+0.096	+0.115
sig.	yes	yes	yes	yes	yes	yes	yes	yes

The 1960s peak and mild post-2000 decline are consistent with the HEADNOTE convention *adding* to the effect, but the effect does not depend on it.

The reasoning's real contribution. Cutting the contaminated tail off the whole-document arm isolates it:

Run	ildc_minus128−masked	Sig.
land forum (842)	$+0.027\ [+0.014, +0.039]$	yes
land temporal (722)	$+0.022\ [-0.0002, +0.043]$	*no*
corpus temporal (2,288)	$+0.023\ [+0.010, +0.035]$	yes
corpus random (2,414)	$+0.029\ [+0.022, +0.036]$	yes

Four point estimates between $+0.022$ and $+0.029$, significant in three; the one failure is the smallest test set, so the earlier non-replication was a power problem, not a contradiction.

15.4 Replication on ILDC itself

Licence access to the IL-TUR release came through, so `src/eval/ildc_ablation.py` ran on ILDC's own documents and labels (`experiments/paper1/ildc_ablation.json`): 31,912 training documents (capped at 8,000 for TF-IDF), 1,516 test documents, test positive rate 0.503. The `facts_only` arm applies this project's ğ5.2 masking to ILDC text; it yields usable text for 1,374 of the 1,516 test documents.

Arm	AUROC	vs facts_only	
facts_only (masked)	0.551	—	
w0 = last 512 words (ILDC's best-model input)	0.607	$+0.055\ [+0.024, +0.086]$	significant
w1 = set back 512	0.571	$+0.018\ [-0.013, +0.050]$	not significant
w2	0.572	$+0.014\ [-0.016, +0.045]$	not significant
w3	0.551	$-0.005\ [-0.038, +0.027]$	not significant
whole document minus its last 128 words	0.577	$+0.024\ [+0.005, +0.042]$	significant

The shape replicates: only the window adjacent to the deleted decision beats facts-only, and w0 beats w1 directly ($+0.038\ [+0.007, +0.070]$). **Three things differ.** (i) The magnitude is a third of ours: $+0.055$ against $+0.18$. Plain TF-IDF is weak on ILDC — 0.55–0.61 AUROC on every arm — so there is less signal to redistribute; this is a statement about the method's reach on this data, not about the leak. (ii) The contamination reaches further: the setback advantage is still significant at 256 words and gone only at 512 (ours: 128), so ILDC's boundary leaves more residue, or residue of a different kind. (iii) The disposition-cue excess in w0 is small ($+0.016$ over the base rate, against $+0.150$ on our corpus), so whatever carries the signal is mostly not the cue patterns this project's regexes know. As on our corpus, the reasoning with the tail removed still adds a little ($+0.024$).

The defensible claim is therefore *qualitative replication on the benchmark itself*: ILDC's best-model input is significantly advantaged by its position next to the deleted decision, and the advantage does not survive moving the window back. The magnitude on ILDC should not be quoted from a TF-IDF model; the fine-tuning pipeline in `ft/` is the instrument that would size it.

15.5 Is the baseline itself clean?

The correction above came from assuming an arm was clean without testing it, so the same question must be put to `masked` — the baseline every comparison runs against and the text extraction reads. There was a concrete reason for concern: this repository contains *two* disposition detectors that disagree by construction. `mask.py`'s `OUTCOME_CUE` requires the proceeding noun *adjacent* to the verb; `labels.py`'s patterns allow a 90-character gap and also cover "the appeal fails" and "we hereby dismiss the appeal".

Arm	AUROC	Δ	Sig.
`masked` (2,647 words)	0.662	—	—
`masked_strict` *(escaped sentences removed)*	0.662	-0.0001 $[-0.0007, +0.0005]$	**no**
`masked_w0` *(its last 512 words)*	0.671	—	—
`masked_w1` *(set back, volume-matched)*	0.644	$+0.027$ $[-0.002, +0.056]$	**no**

Table 10: `experiments/paper1/masked_integrity_forum_heldout.json`.

Sound. Only 109 sentences across 839 cases escape — **one word per case** out of 2,647 — and removing all of them changes AUROC by -0.0001. Reading the list explains why they are harmless: statutory quotations ("that a suit must fail by reason of some formal defect", Order 23 CPC), prior-court dispositions whose forum the narrow regex did not name ("A single judge partly allowed the appeal"), legal propositions and conditionals. The same setback design applied to `masked` gives $+0.027$ with an interval spanning zero at cue rates 0.054/0.031, against $+0.169$ $[+0.136, +0.202]$ at 0.419/0.089 for the contaminated window.

A caveat the test supplied about itself. The naive check — simply deleting `masked`'s last 128 words — *was* significant ($+0.0058$ $[+0.0023, +0.0092]$). That is confounded: deleting 128 words of anything loses information. Only the volume-matched arms distinguish contamination from ordinary loss, and without them this section would have reported a leak that is not there.

The headline

The input construction used by the published state of the art beats a facts-only input by $+0.111$ to $+0.181$ AUROC across four splits. **Essentially all of that is an artifact of imperfect order removal.** Deleting the operative order leaves roughly **128 words** of outcome-telegraphing text in place — a boundary found independently four times — and a word-count-matched window set back past it scores no better than the facts on any split.

With the tail removed, the court's *entire* reasoning adds about $+\mathbf{0.025}$ AUROC. The apparent effect is **four to seven times** its real size. Measured on **26,688** Supreme Court judgments, 1950–2025, every case type, significant in **all eight decades**.

16 The discretisation result

A **representation ladder**: identical classifier, split, labels and 881 test cases at every rung, with only the representation changing.

Rung	Representation	Features	Macro-F1	AUROC
0	majority class	—	0.346	—
1	masked judgment text	50,000	0.608	**0.657**
2	extracted fact text	41,142	0.586	0.614
3	canonical fact atoms	1,313	0.505	**0.517**

Table 11: `experiments/paper1/representation_ladder_forum_heldout.json`.

Extraction costs **0.043**. Canonicalisation costs a further **0.097**, landing at 0.517 — essentially chance. Five independent tests agree:

1. **The ladder**: $0.657 \to 0.614 \to 0.517$.
2. **Is it *this* ontology?** No. Raw text beats both an authored and a corpus-induced vocabulary decisively, and the two vocabularies are indistinguishable from each other:

Comparison	ΔAUROC	95% CI	P
text − induced	$+0.1138$	$[+0.0732, +0.1545]$	0.000
text − ontology	$+0.0887$	$[+0.0430, +0.1352]$	0.000
induced − ontology	$\mathbf{-0.0251}$	$[\mathbf{-0.0711}, \mathbf{+0.0200}]$	**0.846**

3. **Would more labels help?** No. A sweep from 345 to 1,000 entries does not recover the gap, and a frequency-threshold sweep sits at chance throughout.
4. **Where does the loss occur?** The sharpest test. Facts were split into those the vocabulary *can* name and those it cannot: the cost of naming is **0.087** in group A (text 0.544 vs atoms 0.457) and **0.002** in group B (0.526 vs 0.524). The cost is concentrated *where the vocabulary succeeds*, which rules out "coverage is incomplete, fix the vocabulary". Small samples ($n = 143$ and 127); group A's interval spans zero at $P = 0.083$, so this line is suggestive on its own.
5. **A pre-registered prediction**: an LLM given canonical facts would do worse than the same LLM given raw text. It did: macro-F1 0.469 vs 0.491.

The downstream consequence is ğ10: 0 of 498 patterns survive correction.

17 The intended architecture, built properly and measured (v2)

Everything above measures a pipeline that was *specified* one way and *implemented* another. The intended design — the professor's framing, and the project's original plan — runs:

facts → fact families → fact patterns by claim family → **elements** → claims and defences → issues → win/lose

with a second input stream from **authorities** (statutes, precedents) supplying the rules that define each element. ğ16 appeared to condemn that design. It did not, because v1 got two things wrong about it, and ğ17 is the rebuild. The code lives in `v2/`, additive and importing from `src/` rather than duplicating it; `v2/ARCHITECTURE.md` is its plan.

Error 1 — step 2 was built in its lossiest form. The design says *"fact family — classification model run on all embeddings of the facts"*. v1 instead hard-mapped each fact onto one of 345 authored labels by `argmax`.

Error 2 — every layer was scored on win/lose. Which ğ13 shows is near unlearnable here. The visible consequence is that v1's model runs with `features_per_group = {F, S, R, C, Q}`: groups `P` and `E`, the two middle stages of the intended design, are both **empty**. In v2 every stage is scored on its own task.

17.1 S2 — soft fact families

Keeping the embeddings and assigning softly, with no `argmax`. Same 885 test cases, $k = 345$:

Representation	AUROC	dims
masked text	0.666	50,000
soft histogram, temperature 0.25–0.5	**0.633**	**345**
extracted fact text	0.629	47,354
hard `argmax`, same centroids	0.592	345
embedding mean	0.558	4,096
canonical atoms (v1's choice)	0.517	345

Table 12: `v2/experiments/s2_soft_families_forum_heldout.json`.

Step 2 done right loses essentially nothing: 345 dimensions match the 47,354-feature fact text they derive from, and +0.116 over v1. A uniform blur scores 0.512, confirming the representation carries nothing structural by construction, and AUROC declines monotonically as the temperature sharpens toward the `argmax`.

A correction, recorded rather than quietly fixed. The first write-up of this result claimed "the `argmax` alone costs +0.084 AUROC". That was an over-attribution of exactly the single-cause kind this project kept making: the +0.084 compared a soft histogram against v1's canonical atoms, which differ in *three* ways at once — soft versus hard, corpus-induced centroids versus authored labels, and histogram versus binary encoding. Controlled properly, with identical centroids and encoding and only soft-versus-hard varying, the `argmax` costs +0.040, 95% CI $[-0.0015, +0.0826]$: **marginally not significant**. The +0.116 is real and remains the largest single improvement available, but it is a combination, not one mechanism.

17.2 S4 — the element layer

v1 skipped this entirely because the burden metadata it wanted does not exist for any of the 66 elements (B2, ğ21), leaving the architecture's central claim unmeasured. v2 builds it unweighted: one call per (case, claim), a three-way verdict per element, and a **verbatim quote** that is mechanically checked against the facts shown. A verdict whose quote cannot be located is discarded and counted as a refusal, which yields a fabrication metric needing *no ground truth*.

Full run: 4,674 cases × top-2 claims = 9,348 calls, 102 minutes. **40,078 gated verdicts** — 13,008 SATISFIED (32%), 1,481 NOT_SATISFIED (3.7%), 25,589 UNCLEAR (64%). **Fabrication 0.62%** (206 quotes absent, 44 too short). Zero never-answered, zero schema failures.

The model had to change, and establishing why took three tests. `llama4-scout-17b` — v1's default everywhere, chosen on an extraction benchmark where it matched models 6.8× slower — emits **zero** NOT_SATISFIED. A layer that cannot say that cannot express the "or defeat" half of the design, so this was blocking rather than cosmetic.

Condition	SAT	NOT_SAT	UNCLEAR	fabricated
`scout`	15	**0**	150	0.6%
`scout`, prompt rewritten to solicit negatives	23	**0**	151	0.6%
`scout`, shown the court's reasoning (leaky diagnostic)	21	**0**	149	2.9%
`qwen3-235b-a22b-instruct-2507`	27	**4**	56	**0.0%**

Two hypotheses were refuted before the third was confirmed: *prompt over-correction* (rewriting it to call the verdict "common in contested litigation" with five examples changed nothing), and *defeats live in the reasoning that ğ5 removes* (showing the model that text still gave zero). It is the model — and the larger one fabricates *less*. **A model choice validated for extraction does not transfer to judgement tasks.**

The verdicts are legally sensible. The clearest NOT_SATISFIED produced: *"Sham Lal was a tenant of a room in property unit No. B-VI-33"*, defeating an adverse-possession element — tenancy is permissive possession, so it negates hostility. That is the inference the architecture exists to make, and it arrived unprompted.

An accidental reliability study. Two S4 processes ran concurrently — a launch wrongly believed to have failed, plus its replacement — and both wrote the same file: 18,696 rows for 9,348 jobs, exactly 2× every key. **This is the "driver ran two stages at once" defect of ğ19 recurring**, and the lesson is the same: verify a background job started rather than inferring it from absent output. Deduplicated on `(case_id, claim_id)`, the accident bought a free test–retest study at temperature 0:

element verdicts comparable across both runs	40,000
agree	**38,752 (96.9%)**
disagree	1,248 (3.1%)
SATISFIED ↔ UNCLEAR	854
NOT_SATISFIED ↔ UNCLEAR	237
SATISFIED ↔ NOT_SATISFIED	**157 (0.4%) — sign flips**
identical quote when both agree on a non-UNCLEAR verdict	92.7%

96.9% exceeds v1's 87.0% rules-versus-LLM label agreement. It also establishes that **temperature 0 with a fixed seed is not deterministic on this endpoint**, which matters before any result here is described as exactly reproducible.

17.3 The acceptance test, and why S4 fails it

The architecture's central claim — that elements are the right intermediate representation — is falsifiable without gold: *do element features predict the outcome better than the S2 soft fact representation they are derived from?* S2 is the honest baseline rather than v1's atoms, since both come from the same facts, so a gain is attributable to the element abstraction and not to better preprocessing.

Arm	AUROC	features
S2 soft facts	**0.633**	345
S4 elements	**0.521**	70
S2 + S4	0.616	415

Comparison	Δ	95% CI
S4 − S2	−0.1125	$[-0.1625, -0.0639]$
S2+S4 − S2	−0.0169	$[-0.0345, +0.0011]$

Not a null — significantly worse. The element layer destroys most of the signal in the facts it derives from, and adding it to S2 does not help.

Three explanations, two of them mine and both refuted. All testable for free, since every fact carries a cached embedding. Same matched 730 cases:

Arm	AUROC	features
masked text (ceiling)	0.659	50,000
S2 soft facts	0.617	345
S4, old $\pm 1/0$ encoding	0.531	66
S4, *H1*: asked-and-unclear separated from never-asked	0.529	264
S4, *H2*: **soft** element similarity, no LLM at all	0.524	66

H1 — UNCLEAR and never-asked were both encoded 0, conflating two different facts about the world. True, and worth −0.003. *H2* — S4 hard-discretises into three labels, repeating S2's own mistake one level up; the analogue of S2's winning move is a continuous 66-dimensional element-similarity vector straight from the embeddings. It scores 0.524, **no better than the hard version**. Softening bought +0.116 at the fact level and *nothing* at the element level.

The matched-granularity control that settles it. The last confound was dimensionality, 345 against 66. So S2 was re-run at $k = 66$:

66-dimensional basis, soft, same embeddings, same cases	AUROC
induced centroids (data-driven)	**0.632**
authored legal elements (doctrine)	**0.524**

And induced-66 (0.632) equals induced-345 (0.633), so **granularity is irrelevant** and the confound is dead. S4 fails on neither channel width, nor quantisation, nor encoding, but on the **basis**.

A flaw in the acceptance test, stated. Masked text reaches 0.659 and S2 reaches 0.617, so **any fact-derived layer has at most 0.043 to gain over S2**. "Beat S2" was close to unwinnable and the test should have said so. It does not excuse 0.524, which is far *below* S2 rather than merely failing to exceed it, but the bar was badly set.

17.4 Naming the induced clusters — the finding that explains all of it

If the data-driven basis beats the doctrinal one, perhaps the clusters *are* a better ontology. All 66 were named by `qwen3-235b` from their twelve nearest facts, with a `kind` field and a prompt written to resist the flattering answer — it states that an honest `procedural` label is worth more here than a legal-sounding one. 66 of 66 named.

kind	all 66	top 20 by train-only strength
legal_substance	27 (41%)	8 (40%)
procedural	19	8
temporal_quantum	18	4
party_entity	2	0

Kind does not predict predictiveness. Restricting the 66-dimensional histogram to each kind:

Basis	AUROC	dims
all 66 clusters	0.632	66
procedural only	0.609	19
legal_substance only	0.606	27
temporal_quantum only	0.604	18
all *minus* procedural	0.622	47

The finding that explains the whole sequence

Every subset lands at 0.604–0.632. Legal substance alone, court machinery alone, and dates and sums alone are *indistinguishable*. **There is no concentrated outcome signal in these facts. It is weak and smeared across the entire fact space, so any reasonably complete projection captures most of it and no projection captures much.**

That single fact explains everything above: why S2 scores ≈ 0.63 at both $k = 66$ and $k = 345$ (both *span* the space); why the authored elements reach only 0.524 (66 doctrinal elements are a *selective probe* — about 8 asked per case, each a specific legal proposition — not a spanning basis, so they miss a diffuse signal); why softening the element layer bought nothing; and why the headroom above S2 is only 0.043.

So there is no ontology that unlocks this, because there is no concentrated structure to carve. The induced clusters are not a better ontology waiting to be read off — they win by *covering* the fact space.

What the clusters do say: a dispute-regime regularity. The content is coherent even though it is not elemental. The LOSE-direction legal clusters are almost all one thing:

Cluster	Favours	Example member fact
`land_reforms_legislation_enactment`	LOSE	"The West Bengal Land Reforms Act, 1955 was introduced"
`vesting_of_proprietary_rights`	LOSE	"Proprietary rights in sir and khud-kashat land … vested in the State"
`land_vesting_in_state`	LOSE	
`accession_of_princely_states`	LOSE	
`religious_endowment_and_institution`	LOSE	
`document_execution_and_admission`	**WIN**	
`contract_performance_failure`	**WIN**	

Post-independence land-reform and state-vesting cases lose; private document and contract disputes win. That is a finding about Indian land jurisprudence — a *subject-matter regime* regularity, not element satisfaction — and it is precisely why an element ontology could not capture it. Elements ask "is hostility established?"; the data says "is this a zamindari abolition matter against the State, or a private sale-deed dispute?"

A masking defect found on the way. The strongest single cluster, `dismissal_of_appeal_or_petition`, has member facts reading literally "The appeal was dismissed". `mask.py`'s `OUTCOME_CUE` matches `is|are|stands|shall stand` and **not the past tense**, while `PRIOR_COURT_DISPOSITION` requires a named forum. So all of these escape *both* detectors: "The appeal was dismissed", "The suit was dismissed on the ground of limitation", "The Munsif dismissed the suit". It favours **WIN**, which is the signature of a *prior-court* disposition rather than leakage of this court's order — the party lost below, obtained leave, and succeeded here — so it is group `Q`'s procedural shortcut reappearing inside the fact stream. The decomposition above bounds the damage at **0.010** AUROC, so it drives no reported result, but `OUTCOME_CUE` should gain past-tense forms and `PRIOR_COURT_DISPOSITION` a wider forum list.

17.5 Is S4 right? An AI-reviewed gold set, and where the defeat verdicts come from

S4 fails as a feature layer but its verdicts were never checked. No human reviewer is available, so the 40-case, 214-judgement set was annotated by an **AI reviewer** (`claude-opus-5-5`), **blind**: S4's verdicts were stripped before any case was read, and the annotations are stored under an annotator name that says so (`Data/gold/element_gold_annotations_ai.jsonl`). This measures agreement between two models, not legal correctness; B6 remains open.

The first set measured a mismatch. The task builder took each case's facts from an earlier extraction rather than the facts S4 judged. Only 49 of S4's 138 quotes were visible to the reviewer, and agreement was 76% where the evidence matched against 48% where it did not. The set was rebuilt from S4's own facts (same cases, same items, 138/138 quotes visible) and re-annotated.

	SATISFIED P / R	**NOT_SATISFIED** P / R	**overall**
mismatched facts (withdrawn)	72 / 73%	28 / 67%	63.6%
S4's own facts	**80 / 82%**	**37 / 67%**	**71.5%** [65, 77]

Table 13: `v2/experiments/s4_ai_reviewer_agreement.json`. 214 judgements; S4's disagreement with *itself* across two runs is 3.1%, so the remaining 28.5% is not instability.

Positive verdicts are usable; defeat verdicts are not. The reviewer agreed with 16 of S4's 43 `NOT_SATISFIED`, and in 22 of the other 27 judged the element `UNCLEAR`: S4 reads silence, or a nearby procedural fact, as a contradiction.

What the defeat verdicts rest on. Classifying the fact behind each of S4's 138 decisive verdicts (single AI coder, `s4_cited_fact_types.json`): 92 primary facts, 24 conclusions stated as bare facts, 12 court findings, 10 party assertions. Conclusions and findings back **20 of the 43**

NOT_SATISFIED verdicts, and agreement on finding-backed verdicts is 3/12. The usual mechanism is extraction dropping the attribution: "it was concluded that … the plaintiff did not acquire any good or valid title" becomes the fact "The plaintiff did not acquire valid title". Masking cannot catch it — a prior court's view legitimately belongs in masked_text; only the label saying whose view it is was lost.

How common, and does removing them help?. v2/attribution_audit.py flags facts whose source sentence carries a court-attribution cue the fact text dropped: 1,883 of 96,623 (1.9%), in 23% of cases, hand-checked precision 38/51 = 0.75. Re-running S4 on the 1,126 (case, claim) pairs that ever produced a NOT_SATISFIED, plus the gold cases, with those facts excluded or tagged PRIOR COURT FINDING under a prompt forbidding their use as evidence:

	NOT_SAT	finding-backed survive	evidence-backed survive	gold NOT prec.
original	1,481	—	—	16/43 (37%)
exclude	1,279	36.6%	86.8%	14/34 (41%)
tag	1,101	47.6%	72.7%	15/32 (47%)

Table 14: v2/experiments/s4_findings_rerun.json. Gold agreement rises 71.5% → 73.7% → 75.1%; intervals overlap at this *n*. The reviewer saw the findings, so the comparison slightly favours the original.

The fix helps, but the cue detector reaches only a slice: corpus-wide, just 82 of 1,481 NOT_SATISFIED quote a detected finding, against 20/43 in the hand-classified gold set. The hand class included conclusions with no attribution cue at all ("the respondent terminated the tenancy by means of a valid notice"), and the gold set over-samples defeat-bearing cases. Most weak defeat verdicts are not attribution-stripped findings; they are characterisations a cue cannot see.

A classifier instead of a cue: what kind of statement is each fact?. v2/fact_types.py labels every fact PRIMARY, PROCEDURAL, FINDING, LAW or ASSERTION, reading the fact's *source sentence* rather than its text (qwen3-235b, 15 facts per call, 8,620 calls, 8 skips relabelled). Piloted first on the gold cases against the hand labels: finding-or-law recall 30/36, precision 30/41. Over the corpus: PRIMARY 51.8%, FINDING 15.3%, PROCEDURAL 14.5%, ASSERTION 10.8%, LAW 7.6%.

This restores the gold-set estimate at corpus scale. Of all 1,481 NOT_SATISFIED verdicts in the full S4 run, **48.8% quote a fact labelled** FINDING, against a 15.3% base rate; SATISFIED verdicts cite findings at roughly the base rate (16.3%) (v2/experiments/s4_verdicts_by_fact_type.json). The cue detector's 82 was its own blind spot, not the true rate. S4's defeat half is largely a restatement of courts' conclusions.

Two more arms on the same pairs:

	NOT_SAT	of which quote FINDING/L
original	1,481	766 (5
typed: FINDING/LAW tagged, prompt v4 forbids quoting them	1,094	589 (5
exclude_typed: FINDING/LAW removed	590	

The model does not obey the tag for defeat verdicts: under a prompt that forbids it, 54% of its `NOT_SATISFIED` still quote a tagged finding, against 16% of its `SATISFIED`. Labelling is not a control on this model. **Removing the findings halves the defeat verdicts** (1,481 → 590): roughly 60% of S4's defeats exist only because a court's conclusion was in the fact list. Gold agreement *falls* under removal, but that comparison cannot adjudicate: the AI reviewer treated findings as evidence too, so the reference rewards verdicts built on them, and the classifier's 73% precision means some genuine primary facts were removed with them. What the evidence-only defeat layer is worth needs a reference built on primary facts alone.

For the demo and any presentation, the `tag` arm is used: best agreement with the reviewer (75.1%), no extra cost, and defeat verdicts marked low-confidence.

17.6 Where everything ranks

Every non-leaky arm in either version, best first. Case sets differ slightly where noted; the `masked`-versus-S2 comparison is paired on 730 identical cases at +0.0429, CI [+0.0089, +0.0752].

	Arm	AUROC	features	Version
0	**masked text, fine-tuned InLegalBERT (3 seeds)**	**0.695**	—	ft
1	**masked text, plain TF-IDF**	**0.666**	50,000	v1
2	S2 soft fact families	0.633	345	**v2**
3	extracted fact text	0.629	47,354	v1
4	`gbm C only` (metadata alone)	0.612	38	v1
5	`lr S only` (statutes alone)	0.611	49	v1
6	`gbm F+P+E+S+R+C` (full engineered stack)	0.609	1,415	v1
7	canonical atoms	0.536	1,313	v1
8	**S4 elements**	**0.521**	70	**v2**
per-decade majority baseline: macro-F1 0.554, against S2's 0.612				

Two readings matter. **S2 matches v1's entire 1,415-feature engineered pipeline using 345 dimensions and no side inputs at all** — no statutes, no precedents, no metadata — which is a real win on parsimony. And **reading the raw text still beats every structured representation**: plain TF-IDF significantly, and a fine-tuned transformer by a further, smaller margin (ğ17.7). After both versions, no intermediate representation in the architecture predicts better than the text it was derived from.

17.7 Can the published 78% be beaten?

The question splits in two and only one half is worth answering.

On the published task — reasoning included — we already reach **77.6% accuracy** with TF-IDF and logistic regression against the reported 78%, and a fine-tuned transformer reading the same window would very likely clear it. **That result would be worthless**, because ğ15.2 shows the window is essentially all leftover disposition language. Beating 78% that way means getting better at reading the answer off the page.

On the honest task — facts and arguments only — the best arm reaches **62.1% accuracy**, sixteen points below. Six independent lines say that gap is close to irreducible: the masked-text ceiling of 0.659–0.666 across four splits and 26,688 judgments; every structured model at or below a per-decade majority baseline; five LLM configurations likewise; model and LLM

errors statistically *independent* (ğ13); all kinds of fact scoring ≈ 0.605 (ğ17.4); and only 0.043 of headroom above S2.

The one legitimate gap, closed: fine-tuning. The objection that only TF-IDF was tried is now answered. `ft/` fine-tunes InLegalBERT (pretrained on Indian legal text) on the masked arm, case-level by averaging 512-token chunks, on the same 901 `forum_heldout` test cases, with the experiment and its decision bands pre-registered in `ft/PLAN.md` before any run.

Arm	AUROC	Accu
TF-IDF, refit on the identical case set	0.667	6
InLegalBERT, seed 0 / 1 / 2	0.701 / 0.693 / 0.691	66.5 / 64.9 / 6
InLegalBERT, mean of three seeds	**0.695** (sd 0.005)	**64.6%** (sd
capability control: same model on the contaminated last-512 window	*0.907*	*8*

Table 15: `ft/experiments/ft_masked_summary.json`. Paired bootstrap on the same cases: seed 1 − TF-IDF $+0.026$ $[-0.002, +0.056]$; seed 2 $+0.024$ $[-0.005, +0.053]$; the average of seeds 1 and 2 $+0.031$ $[+0.003, +0.059]$. Seed 0's per-case probabilities were not saved.

Three readings. **The gain is small and only marginally established**: no single seed is significantly better than word counts, and the two-seed average is. The mean sits 0.005 above the pre-registered 0.69 boundary, so the band is "ceiling softens" — narrowly. **The ceiling moves from about 0.67 to about 0.70, not to 0.75**, which is inside the 2–6 points predicted above and strengthens rather than weakens the ceiling claim. And **the capability control reproduces the leakage result with a strong model**: the same network, training and cases reach 0.907 on the contaminated window against 0.695 on facts, a gap of $+0.21$, larger than TF-IDF's $+0.18$. The leakage finding is not an artifact of a weak model. Accuracy is too noisy across seeds (62.5–66.5%) to claim anything from.

The comparison to refuse

Never place this project's honest number beside the published 78%. A reader sees 0.68 against 0.78 and concludes we did worse, when the two measure different things and ours is strictly harder. That comparison is the exact error this project exists to expose.

State both with the gap as the result: *on the published input construction we reach 77.6%, essentially reproducing the reported 78%; on facts and arguments alone the same apparatus reaches 62.1%. The fifteen-point gap is the measurement.*

That experiment is the one above: a fine-tuned transformer on masked text reached 0.695, not 0.75, so the ceiling claim survives the objection it was most exposed to.

18 Prior art, and what is actually ours

Both headline findings independently rediscover active, recently-published work. Three papers were located and **verified by fetching the abstract**, not taken from a search summary.

Our finding	Prior art
Facts-only $\ll$ last-512 on Indian SC judgments	**Nigam & Deroy**, *Fact-based Court Judgment Prediction*, arXiv:2311.13350 (Nov 2023). Runs facts-only on *ILDC itself* and reports it worse than the full-document setup. Our direction, on the actual benchmark, three years earlier.
The gap is leakage / shortcut learning	**Watson et al.**, arXiv:2607.04261 (Jul 2026). 33,158 UK Employment Tribunal claims, nine authors. Identifies leakage by *human judgment stratification* and lexical feature flagging; trains a leakage-only model on 200 trigrams.
Ontologising legal text destroys downstream signal	**Sadowski & Chudziak**, arXiv:2608.12326 (May 2026). Six LLMs, three ontology-learning methods.

Unverified leads, to check before citing: ECtHR work reporting that masking verdict words was "not completely successful"; *Multi-Legal-Bench* on label leakage (arXiv:2605.29738); Medvedeva et al., *Rethinking the field of automatic prediction of court decisions* (Springer AI & Law 2021).

18.1 What survives as ours

1. **The volume-controlled positional setback curve.** Every prior method located identifies leakage *lexically* or by *human annotation*. Ours needs no cue detector, no annotator and no feature flagging — only word-count-matched windows at varying distance.
2. **Evidence that this matters, not merely differs.** Our own cue audit **failed to replicate in three of four runs** ($+0.150$, -0.058, $+0.018$, ≈ 0) while the positional curve held in all four with the same boundary. So the claim is not "we have a different method" but *the method the field uses to localise leakage is unstable across case mixes, and here is a cue-free one that is not* — shown on 26,688 judgments with a failed lexical arm inside every run.
3. **The $\sim$128-word margin**, found four times. No located work states a distance. This is the actionable output: a benchmark built this way needs that margin, not zero. It is a lower bound under one baseline, not a constant — against `ildc_facts_only` rather than `masked` the substitute run put it at 512 with 128 and 256 marginal, so quote the sweep table and name the baseline.
4. **The induced-versus-authored control.** Sadowski & Chudziak explicitly conclude their loss is "*not simply from vocabulary discretization alone*" and do not compare an induced vocabulary against a hand-authored one. We do, at $P = 0.846$. **This is the gap in our favour and the less crowded area**; it should lead the paper.
5. **A cleared baseline** (ğ15.5). Prior work does not verify its own clean arm with the same instrument.

18.2 Are we "breaking the industry"? No

- **The subfield already knows.** Three verified papers across 2023–2026 cover both directions; one is three months old.

- **Deployed outcome prediction cannot have this bug.** Litigation analytics predicts from judge, court, party and case-type features — it *cannot* read the court's reasoning, because at prediction time nobody has written it. The leakage is purely an artifact of retrospectively feeding decided judgments to a model: a **benchmark-construction finding, not an industry one**.
- **Scale limits the claims.** Every outcome-model CI spans about 0.11.

What this is instead: a well-controlled workshop paper (NLLP at EMNLP, or JURIX) with two real methodological contributions and a constructive positive result. A good outcome for a class project; not a disruption.

19 Development record: the defects that mattered

Most of the effort went not into building the pipeline but into establishing that its numbers meant what they appeared to mean. Each defect below would have produced a confident, publishable, wrong answer.

Defect	What it would have caused	How it was caught
`unzip` silently aborting two archives on invalid UTF-8 filenames	12,422 of 26,688 judgments, exit code 0	Counting the result
Normalised-title PDF join stripping the year from every title	A 0.6% join rate presented as a corpus	An asserted join-rate floor
Segmentation returning one unit for 57.7% of cases	Extraction silently operating on 2.6% of its input	Median-units sanity check
Quote verifier rejecting 42% of *correct* quotes	Grounding rate understated; good extractions discarded	Hand audit of 16 quotes
Consequential dismissals read as the disposition	196 cases labelled with the wrong outcome	Auditing rules-vs-LLM conflicts
Group-whole and time-respect split constraints in conflict	Training on the future to predict the past	**A unit test**
Attribution reading only `asserted_by` (84% `court_narrative`)	Coverage reported as 17% instead of 96%	Inspecting the field distribution
Canonicaliser's self-reported "NEW" rate of 0.7%	A vocabulary declared complete while 16.7% was unnameable	Checking answers, not the flag
Batching with a shared label menu	Label fidelity silently halved	Batch-size sweep (4/8/16)
`canonicalize_one` dead after a prompt-ID change	The *fallback* path returning `None` for everything	Reading the validation path

Defect	What it would have caused	How it was caught
Driver running two stages at once, only the last exit code checked	A failed stage advancing the chain as success	A 26-minute orphaned process
macOS idle sleep mistaken for a hang (78, 194, 232 min)	Three runs killed for no reason	`pmset -g log`
Burden annotation 92% one class, one value never used	Legal metadata that was a constant, read downstream as fact	A degeneracy gate that *refuses*
Order-window detector swallowing whole short documents (59 cases)	A third of all unusable cases, cause unknown	Unusable-case audit
Caption detector eating argument text	**545 of 5,255 units** — about 1.2M words corpus-wide — deleted as "caption"	Reading what was dropped
Loop-variable shadowing in the metrics dump	Metrics overwritten by a loop counter	Implausible output value
Per-item LLM loop with no retries (twice)	An outage reported as "the corpus does not state these elements"	`ok=False` $\neq$ empty answer
A closing summary hard-coding "On ILDC's own data"	A substitute result labelled as a real one	The smoke test
Two S4 processes writing one file (the ğ19 concurrency defect, recurring)	18,696 rows for 9,348 jobs; every count doubled	Row count against job count
`OUTCOME_CUE` matching only present tense	"The appeal was dismissed" surviving into the fact stream, worth 0.010 AUROC	Reading a cluster's member facts
A model chosen on an extraction benchmark reused for judgement	An element layer structurally unable to say NOT_SATISFIED	Three tests; two hypotheses refuted

19.1 Five transferable lessons

1. **A control is worth more than a method.** The copy control killed a statute result (0.447 $\rightarrow$ an honest 0.173); the popularity control validated a precedent result; the base-rate control dissolved an entire error taxonomy; Benjamini–Hochberg stopped a pattern library shipping; volume-matching stopped a non-existent leak being reported. In every case the control changed the conclusion.
2. **An indirect argument that an objection is unlikely is not a measurement.** The last-512 contamination was dismissed on two true-but-unresponsive grounds before being tested directly. A detector can mark a boundary *correctly* and still leave the preceding sentence.
3. **Distinguish "no answer" from "the answer is nothing".** This single confusion pro-

duced two separate false findings.

4. **Assert your joins.** A pipeline that silently processes 12% of its input is worse than one that crashes.
5. **Elapsed time is not progress, and 0% CPU is not a hang.**

20 Claims held and withdrawn

Belief	What the measurement said
"The last-512 window is clean, so its advantage shows the court's *reasoning* carries the signal."	**Wrong, and this project asserted it before testing it.** The advantage vanishes once the window is set back 128 words, on all four splits. Essentially all of it is residual disposition language; the reasoning's own contribution is about $+0.025$.
"The `argmax` alone costs $+0.084$ AUROC." *(mine, ğ17.1)*	Over-attributed. Controlled properly — same centroids, same encoding, only soft versus hard — it is $+0.040$, CI $[-0.0015, +0.0826]$, not significant. The $+0.116$ gain is a combination of three changes.
"S4 loses because UNCLEAR and never-asked are both encoded zero." *(mine, H1)*	Worth -0.003.
"S4 loses because it hard-discretises, repeating S2's mistake one level up." *(mine, H2)*	A continuous 66-dimensional element vector scores 0.524, no better than the hard version. Softening bought $+0.116$ at the fact level and nothing at the element level.
"The induced clusters will be a better, data-grounded ontology." *(mine, stage 3d)*	The premise dissolved. Every kind of fact scores ≈ 0.605, so the signal is diffuse; the clusters win by *covering* the space, not by carving it better. No ontology unlocks it.
"Defeats live in the court's reasoning, which ğ5 removes." *(mine)*	Refuted: shown the reasoning, the model still emitted zero NOT_SATISFIED. It was the model.
"Extraction preserves the signal in the text."	Loses 0.043 AUROC; canonicalisation a further 0.097.
"The ontology's vocabulary is the problem; an induced one will do better."	Indistinguishable: -0.025, $P = 0.846$. The loss is discretisation itself.
"The procedural shortcut dominates everything."	Weak. `Q only` reaches AUROC 0.498 — chance.
"Max-pooling will fix the dense retrieval arm."	Worse: R@10 0.039 vs 0.053. The deficit is in the facts.
"ILDC leaks the operative order."	It does not. The authors explicitly delete it, and we verified the claim. What it retains is the reasoning — a different and defensible choice.
"Recovering the 1.2M words of wrongly-deleted argument text will improve attribution."	Flat: $0.20 \rightarrow 0.19$. Absolute attributed facts rose about 15%; the rate did not move.
"A better grounding model will recover the missing `Fact` and `Issue` concepts."	Tested across five models. Rules learned under one fire on nothing under another, and the recovered rules were single-example artifacts generalising to zero real sentences.
"A bigger model will produce usable element-burden metadata."	Two models failed the same degeneracy gate; the larger, more instruction-disciplined one also failed.

The last two illustrate a pattern worth naming: on this project **model capability was almost never the binding constraint**. Instruction discipline, evaluation design and data coverage were.

21 What is not established, and open blockers

B2 — element burden metadata does not exist spec ğ8.3. The ontology requires every element to record who bears the burden of proof, to what standard, and when it shifts, and supplies this for **0 of 66**. Two models produced degenerate output (92% and 87% assigned to one class; one standard never used at all, including where it unambiguously applies). A gate refuses to save such annotations, so **feature group `E` is empty** and every result labelled `F+P+E+...` contains no `E`. The fix is a few hours of expert hand annotation (`Data/gold/element_burden_sheet.csv`), not a larger model.

B3 — the vocabulary cannot be frozen spec ğ6.2. 20.3% of 79,496 canonical facts are out of vocabulary against a 5% freeze threshold. Note ğ16 shows improving coverage would not rescue the predictive result.

B4 — no gold annotation set spec ğ6.4. The specification asks for $\geq$150 cases annotated by a legal reviewer with agreement on a 30-case overlap. The **150 cases are prepared** (`Data/gold/gold_tasks.jsc` and an annotation tool exists (`src/annotate/app.py`), but no reviewer has annotated them, so **extraction precision and recall against gold cannot be computed**. The four proxies in use and what each does *not* establish: grounding rate 0.92 bounds fabrication but says nothing about recall; the legal-statement filter bounds one precision failure mode; rules-vs-LLM agreement 87.0% estimates outcome-label noise, not fact quality; the representation ladder measures downstream utility but cannot attribute a loss to a specific extraction error. **This is the single biggest gap for a reviewer, and it is blocked on human time, not compute.**

B5 — the ILDC replication: run (closed). Access was granted and the replication ran (ğ15.4); it reproduces the shape at a third of the magnitude with TF-IDF. The files sit in `Data/ildc/`, gitignored — they are the authors' release under their licence, not ours to redistribute. Open: sizing the effect with a fine-tuned model.

B6 — S4's *correctness* has no human measurement. Two gaps opened by ğ17. First, S4 is reliable (96.9%) and almost never fabricates (0.62%), but whether its verdicts are *right* needs human labels; `v2/gold_elements.py` builds a 40-case, 214-judgement set from B4's prepared cases, blind and with clicked rather than typed quotes, at a reviewer budget of about 94 minutes. **No human reviewer is available**; the set has been annotated by a blind AI reviewer instead (ğ17.5), which measures model–model agreement and is reported as such. A human pass remains the one thing that would turn S4's correctness from plausible into measured. The second gap this paragraph used to record — that nothing was fine-tuned — is closed (ğ17.7).

Other open items.

- **What the cue scrub cannot separate.** `w0_scrubbed` (0.775) and `w0_cue_free` (0.742) both stay above `masked`, so some telegraphing language escapes the patterns. Whether

that remainder is undetected leakage or genuine summative reasoning is unresolved; the setback curve sidesteps the question rather than answering it.

- **Human verification of outcome labels spec ğ5.1.** The check of 200 labels is outstanding; the 562 `UNRESOLVED` conflicts are the highest-information place to start.
- **The attribution ceiling** (ğ7). Cause unknown.
- **Statistical power.** Every outcome-model CI spans about 0.11; the structured model and a per-decade majority baseline cannot be separated at $n = 881$.
- **ğ8 is mis-specified for this corpus**, not merely unmet.
- **Demo**: built (`demo/`), reads only precomputed results; shows element verdicts from the tagged re-run and marks defeat verdicts low-confidence.

22 Reproduction

1. `.venv/bin/pip install -r requirements.txt`; create `.env` with `VOYAGER_API_KEY=<key>`.
2. `./run_downstream.sh` — re-executes itself under `caffeinate -dimsu` so idle sleep cannot stall it, aborts on any stage's non-zero exit, and runs a QA gate after extraction. Every stage is resumable. `make help` lists individual targets.
3. Headline experiments:

```
python -m src.eval.leakage_probe -split forum_heldout
python -m src.eval.reasoning_ablation -split forum_heldout
python -m src.eval.leakage_setback -split forum_heldout
python -m src.eval.masked_integrity -split forum_heldout
python -m src.data.corpus_wide
python -m src.eval.corpus_scale_setback -split temporal
python -m src.eval.representation_ladder -split forum_heldout
python -m src.eval.vocab_ab -split forum_heldout
python -m src.authorities.statute_predict -split forum_heldout
-rebuild-targets
python -m src.authorities.precedent_retrieve -split forum_heldout
python -m src.predict.models -split forum_heldout
python -m src.trace.evaluate -split forum_heldout -model gbm
  -- v2 --
python -m v2.canon_soft -split forum_heldout
python -m v2.element_layer -workers 64 -resume
python -m v2.s4_eval -split forum_heldout
python -m v2.why_s4_fails -split forum_heldout
python -m v2.name_clusters -split forum_heldout -k 66
python -m v2.gold_elements
```

4. Every number in this report is read from a file under `experiments/`. `src/report/figures.py` reads the same files and *raises* rather than substituting a default if one is missing, so a stale figure cannot be produced silently.

23 Repository map

Path	Contents
writeup/	**This report** and LexGraph_Explained.tex, the plain-language companion. The only documentation in the repository.
src/data/	Adapter, labels, masking, segmentation, splits, ontology conversion, corpus_wide.py.
src/extract/	Fact and claim extraction, canonicalisation, vocabulary.
src/eval/	The experiments: leakage probe, reasoning ablation, setback curve, masked integrity, corpus-scale setback, representation ladder, vocabulary A/B and diagnosis, error analysis, ILDC replication.
src/authorities/	Statute normalisation and prediction, precedent retrieval.
src/predict/	Features, outcome models, LLM baselines.
src/trace/	Trace construction and the faithfulness deletion test.
src/llm/client.py	The single door to the model endpoint: cache, timeout, retries.
src/grounding.py	Verbatim quote location with hyphenation repair.
prompts/	Versioned prompt templates, keyed by the IDs the cache records.
ontology/	ontology_v1.yaml and the vocabulary proposals.
experiments/	Every v1 result file. Nothing in this report is sourced elsewhere.
v2/	The intended architecture, rebuilt (ğ17): ARCHITECTURE.md is its plan, with canon_soft (S2), element_catalogue and element_layer (S4), s4_eval, why_s4_fails, name_clusters, and the gold-element annotation app and scorer. Its own experiments/.
tests/	Unit tests, including the split-constraint test that caught a time-travel bug.
Data/interim/	Derived artifacts; large ones are gitignored and rebuild deterministically.
Data/gold/	The 150 prepared gold cases and the 66-element burden sheet (B2, B4).
Data/raw_pdfs/	All 26,688 judgment PDFs (gitignored, 6.8 GB).
run_downstream.sh, Makefile	The driver and individual targets.

This report replaces PROJECT.md, WORKLOG.md and reports/.md, whose history remains in git. Every figure was read from a results file in experiments/.*